

1. Questions

Study the following information carefully and answer the questions given below.

The table given below shows the total number of devices (Laptops + Tablets) sold by four different brands and the ratio of Laptops to Tablets sold.

Brands	Total number of devices sold	Ratio of the number of Laptops to Tablets sold
Samsung	420	4:3
Apple	350	3:4
Sony	600	7:5
Nokia	840	5:7

Note: The total number of devices sold = The number of Laptops sold + The number of Tablets sold

Find the ratio of the total number of Laptops sold by Apple and Sony together to the total number of Tablets sold by Samsung and Nokia together.

- a. 67:55
- b. 50:69
- c. 50:67
- d. 5:7
- e. 25:31

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2. Questions

The total number of Laptops sold by Samsung and Nokia together is what percentage more or less than the total number of Laptops sold by Apple and Sony together?

- a. 15% more
- b. 28% less
- c. 12% less
- d. 18% more
- e. 85% less

3. Questions

Brand Motorola sells 14.28% more Laptops than Nokia and 20% less Tablets than Sony. Find the

total number of devices sold by Motorola.

- a. 500
- b. 600
- c. 650
- d. 580
- e. 610

4. Questions

Find the average number of Tablets sold by Samsung, Apple, and Sony.

- a. 220
- b. 200
- c. 310
- d. 230
- e. 210

5. Questions

If 16.66% of the Laptops sold by Samsung and 12.5% of the Tablets sold by Apple are defective, find the total number of non-defective devices among these two specific categories.

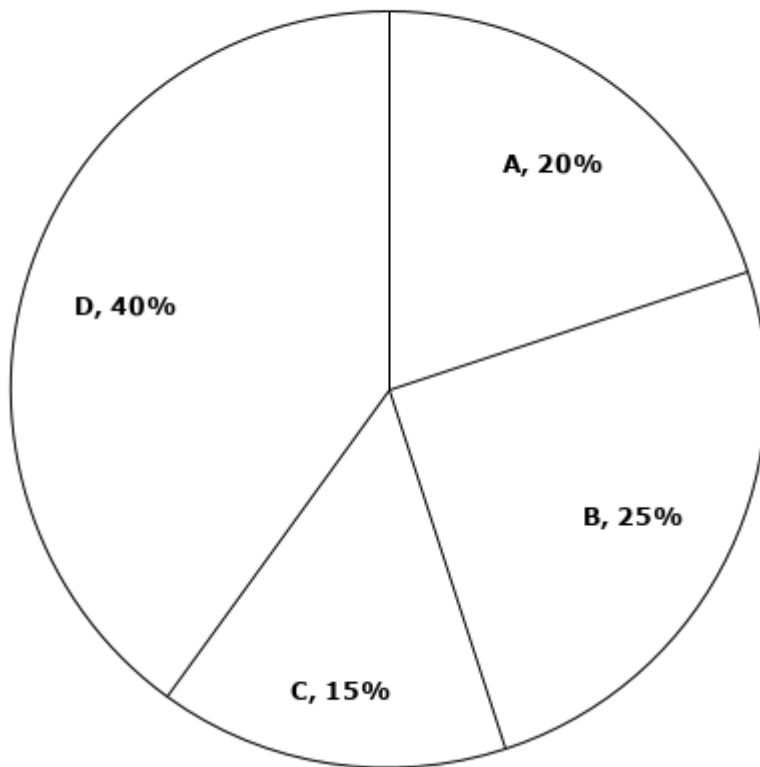
- a. 225
- b. 475
- c. 375
- d. 385
- e. 265

6. Questions

Study the following information carefully and answer the questions given below.

The pie chart given below shows the percentage distribution of the total number of employees working in four different departments (A, B, C, and D) of a corporate office. The total number of employees across all departments together is 2400.

The percentage distribution of the total number of employees working



Notes:

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(i). The total number of employees = The number of female employees + The number of male employees

(ii). The total number of employees = The number of executives + The number of managers

If the ratio of the number of males to females in departments A and C is 5:3 and 7:5, respectively. Find the difference between the number of males in departments A and C together and the number of females in departments A and C together.

- a. 280
- b. 160
- c. 190
- d. 180
- e. 150

7. Questions

Find the central angle corresponding to the total number of employees in departments B and C together.

- a. 154°
- b. 144°
- c. 124°

d. 134°

e. 140°

8. Questions

What is the difference between the highest and the lowest number of employees among the four departments?

a. 540

b. 560

c. 580

d. 600

e. 620

9. Questions

If $\frac{5}{8}$ th of the employees from department B and $\frac{4}{9}$ th of the employees from department C are transferred to form a new department E. Find the total number of employees in department E.

a. 525

b. 535

c. 435

d. 635

e. 545

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10. Questions

The total salary of all employees in department A is Rs. 14.4 lakhs. If the department consists only of managers and executives receiving Rs. 5000 and Rs. 2000 each respectively, find the number of managers.

a. 320

b. 260

c. 180

d. 160

e. 150

11. Questions

A boat's speed in still water is 15 km/hr and the stream's speed is 3 km/hr. If it takes $(2x + 1)$ hours to cover 270 km downstream and 144 km upstream together, find the value of x .

a. 15

- b. 23
- c. 12
- d. 13
- e. 14

12. Questions

Find the difference between the interest earned when ₹8,000 is invested in Scheme A at 15% p.a. simple interest for 4 years and in Scheme B at 20% p.a. compound interest, compounded annually, for 2 years.

- a. Rs. 1380
- b. Rs. 1480
- c. Rs. 1180
- d. Rs. 1220
- e. Rs. 1280

13. Questions

Inlet pipes A and B can fill a tank in 20 hours and $(y + 5)$ hours respectively, while outlet pipe C can empty the tank in 40 hours. If all three pipes together fill 70 liters of a 120-liter tank in 10 hours, find the value of y .

- a. 20
- b. 15
- c. 45
- d. 30
- e. 25

14. Questions

The perimeter of a rectangular floor is equal to the perimeter of a square floor, which is 112 m. If the ratio of the length to the breadth of the rectangular floor is 4:3, find the difference between the costs of flooring the two floors at the rate of Rs. 15 per m^2 .

- a. Rs. 120
- b. Rs. 360
- c. Rs. 340
- d. Rs. 480
- e. Rs. 240

15. Questions

From a 160-liter mixture of orange juice and water in the ratio 5:3, 32 liters are removed and replaced with pure water. What is the quantity of water in the final mixture?

- a. 24 liters
- b. 40 liters
- c. 42 liters
- d. 32 liters
- e. 80 liters

16. Questions

In a partnership, P, Q, and R invest Rs. 12000, Rs. 15000, and Rs. $1000(k + 5)$ for 12, 8, and 6 months, respectively. Find the value of k, if P's share is Rs. 2400 from a total profit of Rs. 6400.

- a. 15
- b. 20
- c. 25
- d. 10
- e. 12

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17. Questions

The calculated average weight of 20 items is 42 kg. If two items weighing 35 kg and 43 kg were misread as 75 kg and 63 kg respectively, what is the actual average weight?

- a. 41 kg
- b. 29 kg
- c. 39 kg
- d. 37 kg
- e. 45 kg

18. Questions

A shopkeeper purchased an article for Rs. 800 and the marked price of the article is Rs. $(m + 300)$. If he allows a discount of 20% on the marked price and still earns a profit of Rs. 160, find the value of m.

- a. 1300
- b. 800
- c. 1000

- d. 900
- e. 1200

19. Questions

A box contains 6 red, 5 yellow, and 4 black pens. If two pens are drawn at random simultaneously, find the probability that both are of the same color.

- a. 34/105
- b. 11/35
- c. 31/105
- d. 41/105
- e. 8/35

20. Questions

5 years ago, the ratio of the ages of A and B was 3:4. 5 years from now, the ratio of their ages will become 4:5. Find the sum of the present ages of A and B.

- a. 60 years
- b. 90 years
- c. 70 years
- d. 80 years
- e. 85 years

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21. Questions

What value will come in place of the question mark (?) in the following question?

$$(55 / 100) * 800 - 16^2 + \sqrt{(1296)} = ? / 3$$

- a. 660
- b. 220
- c. 650
- d. 640
- e. 440

22. Questions

$$(4 / 9) * 1350 + 840 / 6 - \sqrt[3]{(1331)} = ? - 25$$

- a. 704

- b. 729
- c. 754
- d. 744
- e. 764

23. Questions

$$8.5 * 14 + 1250 / 50 - 12^2 = ? - \sqrt{(289)}$$

- a. 0
- b. -17
- c. 27
- d. 34
- e. 17

24. Questions

$$1728/12 + 19 * 7 - (35 / 100) * 600 = ?$$

- a. 487
- b. 67
- c. 77
- d. 57
- e. 144

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25. Questions

$$(15 / 8) * 320 - \sqrt[3]{(512)} + 14^2 = ? + 11 * 9$$

- a. 788
- b. 699
- c. 887
- d. 689
- e. 679

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

$$44.98\% \text{ of } 800.05 + 16.02 * 8.99 - \sqrt{(1520.95)} = ?$$

- a. 465

- b. 504
- c. 425
- d. 485
- e. 405

27. Questions

$$64.99\% \text{ of } (19.98)^2 + 239.95 / 5.05 - \sqrt[3]{(999.8)} = ?$$

- a. 318
- b. 328
- c. 298
- d. 268
- e. 258

28. Questions

$$(6.01)^3 * 2.98 + 1499.95 / 49.98 - 74.99\% \text{ of } 300.02 = ?$$

- a. 473
- b. 413
- c. 903
- d. 433
- e. 453

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29. Questions

$$184.98 / 4.99 + 23.05 * 11.95 - \sqrt{(1088.99)} = ? + 34.95$$

- a. 245
- b. 315
- c. 215
- d. 265
- e. 280

30. Questions

$$85.05\% \text{ of } 600.01 - 18.98 * 7.02 + \sqrt[3]{(1727.95)} = ?$$

- a. 409
- b. 369

- c. 655
- d. 389
- e. 349

31. Questions

What will come in the place of question mark (?) in the following number series.

8, 10, 23, 73, 297, ?

- a. 1481
- b. 1491
- c. 1501
- d. 1495
- e. 1471

32. Questions

15, 22, 33, 49, 71, 100, ?

- a. 137
- b. 147
- c. 135
- d. 127
- e. 129

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33. Questions

4, 5, 13, 40, 104, ?

- a. 219
- b. 229
- c. 239
- d. 225
- e. 209

34. Questions

20, 100, 50, 250, 125, ?

- a. 625
- b. 600

- c. 615
- d. 635
- e. 650

35. Questions

100, 104, 95, 111, 86, ?

- a. 112
- b. 132
- c. 120
- d. 102
- e. 122

36. Questions

Find the wrong term in the given series.

32, 40, 56, 82, 112, 152

- a. 56
- b. 82
- c. 112
- d. 40
- e. 152

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37. Questions

15, 17, 20, 26, 32, 43, 56

- a. 17
- b. 20
- c. 26
- d. 32
- e. 43

38. Questions

400, 396, 389, 362, 313, 192

- a. 389
- b. 362

- c. 313
- d. 396
- e. 192

39. Questions

55, 70, 95, 135, 175, 230

- a. 135
- b. 70
- c. 175
- d. 55
- e. 95

40. Questions

8, 16, 24, 50, 72, 144

- a. 16
- b. 24
- c. 72
- d. 144
- e. 50

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41. Questions

In each of the following questions, two equations are given. You have to solve both equations to find the relation between x and y.

I). $x^2 - 21x + 104 = 0$

II). $y^2 - 27y + 182 = 0$

- a. $x \leq y$
- b. $x \geq y$
- c. $x < y$
- d. $x > y$
- e. $x = y$ or the relationship cannot be established

42. Questions

I). $2x^2 + 15x + 27 = 0$

II). $2y^2 + 19y + 45 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or the relationship cannot be established

43. Questions

I). $x^2 - 6x - 55 = 0$

II). $y^2 - 22y + 117 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or the relationship cannot be established

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44. Questions

I). $x = \sqrt{(784)}$

II). $y^2 - 24y - 112 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or the relationship cannot be established

45. Questions

I). $5x^2 - 18x + 16 = 0$

II). $y^2 - 11y + 28 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$

d. $x \leq y$

e. $x = y$ or the relationship cannot be established

Explanations:

1. Questions

Samsung

The total number of devices sold = 420

The number of Laptops sold = $420 * (4/7) = 240$

The number of Tablets sold = $420 * (3/7) = 180$

Apple

The total number of devices sold = 350

The number of Laptops sold = $350 * (3/7) = 150$

The number of Tablets sold = $350 * (4/7) = 200$

Sony

The total number of devices sold = 600

The number of Laptops sold = $600 * (7/12) = 350$

The number of Tablets sold = $600 * (5/12) = 250$

Nokia

The total number of devices sold = 840

The number of Laptops sold = $840 * (5/12) = 350$

The number of Tablets sold = $840 * (7/12) = 490$

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Brand	Number of Laptops sold	Number of Tablets sold
Samsung	240	180
Apple	150	200
Sony	350	250
Nokia	350	490

Answer: C

The total number of Laptops sold by Apple and Sony = $150 + 350 = 500$

The total number of Tablets sold by Samsung and Nokia = $180 + 490 = 670$

Required ratio = $500:670 = 50:67$

Hence, the correct answer is option C.

2. Questions

Samsung

The total number of devices sold = 420

The number of Laptops sold = $420 * (4/7) = 240$

The number of Tablets sold = $420 * (3/7) = 180$

Apple

The total number of devices sold = 350

The number of Laptops sold = $350 * (3/7) = 150$

The number of Tablets sold = $350 * (4/7) = 200$

Sony

The total number of devices sold = 600

The number of Laptops sold = $600 * (7/12) = 350$

The number of Tablets sold = $600 * (5/12) = 250$

Nokia

The total number of devices sold = 840

The number of Laptops sold = $840 * (5/12) = 350$

The number of Tablets sold = $840 * (7/12) = 490$

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Brand	Number of Laptops sold	Number of Tablets sold
Samsung	240	180
Apple	150	200
Sony	350	250
Nokia	350	490

Answer: D

The total number of Laptops sold by Samsung and Nokia = $240 + 350 = 590$

The total number of Laptops sold by Apple and Sony = $150 + 350 = 500$

Required percentage difference = $[(590 - 500) / 500] * 100$

= $(90 / 500) * 100 = 18\%$ more

Hence, the correct answer is option D.

3. Questions

Samsung

The total number of devices sold = 420

The number of Laptops sold = $420 * (4/7) = 240$

The number of Tablets sold = $420 * (3/7) = 180$

Apple

The total number of devices sold = 350

The number of Laptops sold = $350 * (3/7) = 150$

The number of Tablets sold = $350 * (4/7) = 200$

Sony

The total number of devices sold = 600

The number of Laptops sold = $600 * (7/12) = 350$

The number of Tablets sold = $600 * (5/12) = 250$

Nokia

The total number of devices sold = 840

The number of Laptops sold = $840 * (5/12) = 350$

The number of Tablets sold = $840 * (7/12) = 490$

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Brand	Number of Laptops sold	Number of Tablets sold
Samsung	240	180
Apple	150	200
Sony	350	250
Nokia	350	490

Answer: B

The number of Laptops sold by Motorola = $350 + (1/7 * 350) = 350 + 50 = 400$

The number of Tablets sold by Motorola = $250 - (20\% \text{ of } 250) = 250 - 50 = 200$

The total number of devices sold by Motorola = $400 + 200 = 600$

Hence, the correct answer is option B.

4. Questions

Samsung

The total number of devices sold = 420

The number of Laptops sold = $420 * (4/7) = 240$

The number of Tablets sold = $420 * (3/7) = 180$

Apple

The total number of devices sold = 350

The number of Laptops sold = $350 * (3/7) = 150$

The number of Tablets sold = $350 * (4/7) = 200$

Sony

The total number of devices sold = 600

The number of Laptops sold = $600 * (7/12) = 350$

The number of Tablets sold = $600 * (5/12) = 250$

Nokia

The total number of devices sold = 840

The number of Laptops sold = $840 * (5/12) = 350$

The number of Tablets sold = $840 * (7/12) = 490$

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Brand	Number of Laptops sold	Number of Tablets sold
Samsung	240	180
Apple	150	200
Sony	350	250
Nokia	350	490

Answer: E

The total number of Tablets sold by Samsung, Apple, and Sony = $180 + 200 + 250 = 630$

Required average = $630 / 3 = 210$

Hence, the correct answer is option E.

5. Questions

Samsung

The total number of devices sold = 420

The number of Laptops sold = $420 * (4/7) = 240$

The number of Tablets sold = $420 * (3/7) = 180$

Apple

The total number of devices sold = 350

The number of Laptops sold = $350 * (3/7) = 150$

The number of Tablets sold = $350 * (4/7) = 200$

Sony

The total number of devices sold = 600

The number of Laptops sold = $600 * (7/12) = 350$

The number of Tablets sold = $600 * (5/12) = 250$

Nokia

The total number of devices sold = 840

The number of Laptops sold = $840 * (5/12) = 350$

The number of Tablets sold = $840 * (7/12) = 490$

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Brand	Number of Laptops sold	Number of Tablets sold
Samsung	240	180
Apple	150	200
Sony	350	250
Nokia	350	490

Answer: C

The number of defective Laptops sold by Samsung = $(1/6) * 240 = 40$

The number of non-defective Laptops sold by Samsung = $240 - 40 = 200$

The number of defective Tablets sold by Apple = $(1/8) * 200 = 25$

The number of non-defective Tablets sold by Apple = $200 - 25 = 175$

Total non-defective devices sold = $200 + 175 = 375$

Hence, the correct answer is option C.

6. Questions

Department A

The total number of employees = $2400 * (20/100) = 480$

Department B

The total number of employees = $2400 * (25/100) = 600$

Department C

The total number of employees = $2400 * (15/100) = 360$

Department D

The total number of employees = $2400 * (40/100) = 960$

Department	Number of employees
A	480
B	600
C	360
D	960

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Answer: D

The number of males in department A = $480 * (5/8) = 300$

The number of females in department A = $480 - 300 = 180$

The number of males in department C = $360 * (7/12) = 210$

The number of females in department C = $360 - 210 = 150$

The total number of males in departments A and C together = $300 + 210 = 510$

The total number of females in departments A and C together = $180 + 150 = 330$

Required difference = $510 - 330 = 180$

Hence, the correct answer is option D.

7. Questions

Department A

The total number of employees = $2400 * (20/100) = 480$

Department B

The total number of employees = $2400 * (25/100) = 600$

Department C

The total number of employees = $2400 * (15/100) = 360$

Department D

The total number of employees = $2400 * (40/100) = 960$

Department	Number of employees
A	480
B	600
C	360
D	960

Answer: B

The combined percentage of employees in departments B and C together = $25\% + 15\% = 40\%$

Required central angle = $(40/100) * 360^\circ = 144^\circ$

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Hence, the correct answer is option B.

8. Questions

Department A

The total number of employees = $2400 * (20/100) = 480$

Department B

The total number of employees = $2400 * (25/100) = 600$

Department C

The total number of employees = $2400 * (15/100) = 360$

Department D

The total number of employees = $2400 * (40/100) = 960$

Department	Number of employees
A	480
B	600
C	360
D	960

Answer: D

The highest number of employees is in department D. = 960

The lowest number of employees is in department C. = 360

Required difference = $960 - 360 = 600$

Hence, the correct answer is option D.

9. Questions

Department A

The total number of employees = $2400 * (20/100) = 480$

Department B

The total number of employees = $2400 * (25/100) = 600$

Department C

The total number of employees = $2400 * (15/100) = 360$

Department D

The total number of employees = $2400 * (40/100) = 960$

Department	Number of employees
A	480
B	600
C	360
D	960

Answer: B

The number of employees transferred from department B = $600 * (5/8) = 375$

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The number of employees transferred from department C = $360 * (4/9) = 160$

The total number of employees in department E = $375 + 160 = 535$

Hence, the correct answer is option B.

10. Questions

Department A

The total number of employees = $2400 * (20/100) = 480$

Department B

The total number of employees = $2400 * (25/100) = 600$

Department C

The total number of employees = $2400 * (15/100) = 360$

Department D

The total number of employees = $2400 * (40/100) = 960$

Department	Number of employees
A	480
B	600
C	360
D	960

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Answer: D

Let the number of managers be X and the number of executives be Y.

$$X + Y = 480 \text{ --- (1)}$$

$$5000X + 2000Y = 1440000$$

$$5X + 2Y = 1440 \text{ --- (2)}$$

Multiplying equation (1) by 2 and subtracting from equation (2):

$$3X = 1440 - 960$$

$$= 480$$

$$X = 160$$

The total number of managers in department A is 160.

Hence, the correct answer is option D.

11. Questions

Answer: D

Speed of the boat downstream = $15 + 3 = 18$ km/hr

Speed of the boat upstream = $15 - 3 = 12$ km/hr

Time taken to travel downstream = $270 / 18 = 15$ hours

Time taken to travel upstream = $144 / 12 = 12$ hours

Total time taken for the journey = $15 + 12 = 27$ hours

According to the problem, the total time taken is $(2x + 1)$ hours.

$$2x + 1 = 27$$

$$2x = 26$$

$$x = 13$$

Hence, the correct answer is option D.

12. Questions

Answer: E

Simple Interest from Scheme A = $\text{Principal} * \text{Rate} * \text{Time} / 100$

$$= 8000 * 15 * 4 / 100$$

$$= 8000 * 60 / 100 = \text{Rs. } 4800$$

Amount from Scheme B = $\text{Principal} * (1 + \text{Rate} / 100)^{\text{Time}}$

$$= 8000 * (1 + 20 / 100)^2$$

$$= 8000 * (1.2)^2 = \text{Rs. } 11520$$

Compound Interest from Scheme B = $\text{Amount} - \text{Principal}$

$$= 11520 - 8000 = \text{Rs. } 3520$$

Difference between the interests = $4800 - 3520 = \text{Rs. } 1280$

Hence, the correct answer is option E.

13. Questions

Answer: E

Total capacity of the tank = 120 litres

Efficiency of Pipe A = $120 / 20 = +6$ litres/hr

Efficiency of Pipe C = $120 / 40 = -3$ litres/hr

Let the efficiency of Pipe B be $+E$ litres/hr.

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Combined efficiency of all three pipes = $6 + E - 3 = 3 + E$ litres/hr

They operate together for 10 hours and fill 70 litres.

$$(3 + E) * 10 = 70$$

$$3 + E = 7$$

$$E = 4 \text{ litres/hr}$$

Time taken by Pipe B to fill the tank alone = Total capacity / Efficiency

$$= 120 / 4 = 30 \text{ hours}$$

According to the problem, Pipe B takes $y + 5$ hours.

$$y + 5 = 30$$

$$y = 25$$

Hence, the correct answer is option E.

14. Questions

Answer: E

Let the rectangular floor length and width be $4a$ and $3a$ respectively.

$$112 = 2 * (4a + 3a)$$

$$= 2 * 7a$$

$$= 14a$$

$$a = 8$$

$$\text{Length of the rectangle} = 4 * 8 = 32 \text{ m}$$

$$\text{Breadth of the rectangle} = 3 * 8 = 24 \text{ m}$$

$$\text{Area of the rectangular floor} = 32 * 24 = 768 \text{ m}^2$$

$$\text{Perimeter of the square floor} = 112 \text{ m}$$

$$\text{Side of the square} = 112 / 4 = 28 \text{ m}$$

$$\text{Area of the square floor} = 28 * 28 = 784 \text{ m}^2$$

$$\text{Cost of flooring the rectangular room} = 768 * 15 = \text{Rs. } 11520$$

$$\text{Cost of flooring the square room} = 784 * 15 = \text{Rs. } 11760$$

$$\text{Difference in cost} = 11760 - 11520 = \text{Rs. } 240$$

Hence, the correct answer is option E.

15. Questions

Answer: E

Initial total quantity of the mixture = 160 liters

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Initial quantity of orange juice = $(5 / 8) * 160 = 100$ liters

Initial quantity of water = $(3 / 8) * 160 = 60$ liters

When 32 liters of the mixture is removed:

Quantity of orange juice removed = $(5 / 8) * 32 = 20$ liters

Quantity of water removed = $(3 / 8) * 32 = 12$ liters

Remaining orange juice = $100 - 20 = 80$ liters

Remaining water = $60 - 12 = 48$ liters

New quantity of water = $48 + 32 = 80$ liters

Hence, the correct answer is option E.

16. Questions

Answer: A

Equivalent investment of P = $12000 * 12 = 144000$

Equivalent investment of Q = $15000 * 8 = 120000$

Equivalent investment of R = $(k + 5) * 1000 * 6 = (k + 5) * 6000$

Total profit = Rs. 6400

P's share = Rs. 2400

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The ratio of P's profit to the total profit is $2400 / 6400 = 3 / 8$

Total investment = $144000 + 120000 + (k + 5) * 6000$

= $264000 + (k + 5) * 6000$

$144000 / [264000 + (k + 5) * 6000] = 3 / 8$

$24 / [44 + (k + 5)] = 3 / 8$

$24 / (49 + k) = 3 / 8$

$192 = 3 * (49 + k)$

= $147 + 3k$

$3k = 45$

$k = 15$

Hence, the correct answer is option A.

17. Questions

Answer: C

Incorrect sum of the 20 items = $20 * 42 = 840$ kg

Sum of the incorrectly recorded weights = $75 + 63 = 138$ kg

Sum of the actual weights = $35 + 43 = 78$ kg

Correct sum of the 20 items = $840 + (78 - 138) = 780$ kg

Correct average weight = $780 / 20 = 39$ kg

Hence, the correct answer is option C.

18. Questions

Answer: D

Cost Price of the article = Rs. 800

Profit earned on the article = Rs. 160

Selling Price = $800 + 160 =$ Rs. 960

The discount offered is 20% on the marked price.

Selling Price = Marked Price * $(100 - 20) / 100$

$960 = \text{Marked Price} * 80 / 100$

Marked Price = $960 * 100 / 80$

$= 12 * 100 =$ Rs. 1200

According to the problem, the marked price is Rs. $(m + 300)$.

$m + 300 = 1200$

$m = 900$

Hence, the correct answer is option D.

19. Questions

Answer: C

Total number of pens in the box = $6 + 5 + 4 = 15$ pens

Total number of ways to draw 2 pens = ${}^{15}C_2$

$= (15 * 14) / (2 * 1) = 105$ ways

Favorable outcomes for drawing 2 pens of the same color means drawing either 2 red, 2 yellow, or 2 black pens.

Ways to draw 2 red pens = ${}^6C_2 = (6 * 5) / (2 * 1) = 15$

Ways to draw 2 yellow pens = ${}^5C_2 = (5 * 4) / (2 * 1) = 10$

Ways to draw 2 black pens = ${}^4C_2 = (4 * 3) / (2 * 1) = 6$

Total favorable ways = $15 + 10 + 6 = 31$ ways

Probability = Favorable ways / Total ways = $31 / 105$

Hence, the correct answer is option C.

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20. Questions**Answer: D**

Let the ages of A and B 5 years ago be $3k$ and $4k$ respectively.

Ages 5 years from now:

$$\text{Age of A} = 3k + 5 + 5 = 3k + 10$$

$$\text{Age of B} = 4k + 5 + 5 = 4k + 10$$

The ratio of their ages 5 years from now is 4:5.

$$(3k + 10) / (4k + 10) = 4 / 5$$

$$5 * (3k + 10) = 4 * (4k + 10)$$

$$15k + 50 = 16k + 40$$

$$16k - 15k = 50 - 40$$

$$k = 10$$

$$\text{Present age of A} = 3 * 10 + 5 = 35 \text{ years}$$

$$\text{Present age of B} = 4 * 10 + 5 = 45 \text{ years}$$

$$\text{Sum of their present ages} = 35 + 45 = 80 \text{ years}$$

Hence, the correct answer is option D.

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21. Questions**Answer: A**

$$(55/100) * 800 - 16^2 + \sqrt{(1296)} = ? / 3$$

$$440 - 256 + 36 = ? / 3$$

$$184 + 36 = ? / 3$$

$$220 = ? / 3$$

$$? = 220 * 3$$

$$= 660$$

Hence, the correct answer is option A.

22. Questions**Answer: C**

$$(4 / 9) * 1350 + 840 / 6 - \sqrt[3]{(1331)} = ? - 25$$

$$600 + 140 - 11 = ? - 25$$

$$740 - 11 = ? - 25$$

$$729 = ? - 25$$

$$? = 729 + 25$$

$$= 754$$

Hence, the correct answer is option C.

23. Questions

Answer: E

$$8.5 * 14 + 1250 / 50 - 12^2 = ? - \sqrt{(289)}$$

$$119 + 25 - 144 = ? - 17$$

$$144 - 144 = ? - 17$$

$$0 = ? - 17$$

$$? = 0 + 17$$

$$= 17$$

Hence, the correct answer is option E.

24. Questions

Answer: B

$$1728/12 + 19 * 7 - (35 / 100) * 600 = ?$$

$$144 + 133 - 210 = ?$$

$$277 - 210 = ?$$

$$? = 67$$

Hence, the correct answer is option B.

25. Questions

Answer: D

$$(15 / 8) * 320 - \sqrt[3]{(512)} + 14^2 = ? + 11 * 9$$

$$600 - 8 + 196 = ? + 99$$

$$592 + 196 = ? + 99$$

$$788 = ? + 99$$

$$? = 788 - 99$$

$$= 689$$

Hence, the correct answer is option D.

26. Questions

Answer: A

$$44.98\% \text{ of } 800.05 + 16.02 * 8.99 - \sqrt{(1520.95)} = ?$$

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$$45\% * 800 + 16 * 9 - \sqrt{(1521)} = ?$$

$$360 + 144 - 39 = ?$$

$$504 - 39 = ?$$

$$? = 465$$

Hence, the correct answer is option A.

27. Questions

Answer: C

$$64.99\% \text{ of } (19.98)^2 + 239.95 / 5.05 - \sqrt[3]{(999.8)} = ?$$

$$65\% * 20^2 + 240 / 5 - \sqrt[3]{(1000)} = ?$$

$$65\% * 400 + 48 - 10 = ?$$

$$260 + 48 - 10 = ?$$

$$308 - 10 = ?$$

$$? = 298$$

Hence, the correct answer is option C.

28. Questions

Answer: E

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$$(6.01)^3 * 2.98 + 1499.95 / 49.98 - 74.99\% \text{ of } 300.02 = ?$$

$$6^3 * 3 + 1500 / 50 - 75\% \text{ of } 300 = ?$$

$$216 * 3 + 30 - 225 = ?$$

$$648 + 30 - 225 = ?$$

$$678 - 225 = ?$$

$$? = 453$$

Hence, the correct answer is option E.

29. Questions

Answer: A

$$184.98 / 4.99 + 23.05 * 11.95 - \sqrt{(1088.99)} = ? + 34.95$$

$$185 / 5 + 23 * 12 - \sqrt{(1089)} = ? + 35$$

$$37 + 276 - 33 = ? + 35$$

$$313 - 33 = ? + 35$$

$$280 = ? + 35$$

$$? = 280 - 35$$

$$= 245$$

Hence, the correct answer is option A.

30. Questions

Answer: D

$$85\% * 600 - 19 * 7 + \sqrt[3]{(1728)} = ?$$

$$510 - 133 + 12 = ?$$

$$377 + 12 = ?$$

$$? = 389$$

Hence, the correct answer is option D.

31. Questions

Answer: B

$$8 * 1 + 2 = 10$$

$$10 * 2 + 3 = 23$$

$$23 * 3 + 4 = 73$$

$$73 * 4 + 5 = 297$$

$$297 * 5 + 6 = 1491$$

Logic: The series follows a pattern of multiplying by consecutive natural numbers and adding the next consecutive natural number.

Hence, the correct answer is option B.

32. Questions

Answer: A

$$15 + 7 = 22$$

$$22 + 11 = 33$$

$$33 + 16 = 49$$

$$49 + 22 = 71$$

$$71 + 29 = 100$$

$$100 + 37 = 137$$

Logic: The double difference of the series follows consecutive natural numbers (4, 5, 6, 7, 8).

Hence, the correct answer is option A.

33. Questions

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Answer: B

$$4 + 1^3 = 5$$

$$5 + 2^3 = 13$$

$$13 + 3^3 = 40$$

$$40 + 4^3 = 104$$

$$104 + 5^3 = 229$$

Logic: The difference between consecutive numbers is the cube of consecutive natural numbers.

Hence, the correct answer is option B.

34. Questions

Answer: A

$$20 * 5 = 100$$

$$100 / 2 = 50$$

$$50 * 5 = 250$$

$$250 / 2 = 125$$

$$125 * 5 = 625$$

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Logic: The series alternates between multiplying by 5 and dividing by 2.

Hence, the correct answer is option A.

35. Questions

Answer: E

$$100 + 2^2 = 104$$

$$104 - 3^2 = 95$$

$$95 + 4^2 = 111$$

$$111 - 5^2 = 86$$

$$86 + 6^2 = 122$$

Logic: The series follows an alternating pattern of adding and subtracting the squares of consecutive natural numbers starting from 2.

Hence, the correct answer is option E.

36. Questions

Answer: B

$$32 + 8 = 40$$

$$40 + 16 = 56$$

$$56 + 24 = 80 \text{ (Therefore, 82 is wrong and should be 80)}$$

$$80 + 32 = 112$$

$$112 + 40 = 152$$

Logic: The series follows the addition of consecutive multiples of 8.

Hence, the correct answer is option B.

37. Questions

Answer: C

$$15 + 2 = 17$$

$$17 + 3 = 20$$

$$20 + 5 = 25 \text{ (Therefore, 26 is wrong and should be 25)}$$

$$25 + 7 = 32$$

$$32 + 11 = 43$$

$$43 + 13 = 56$$

Logic: The difference between consecutive numbers follows the series of consecutive prime numbers.

Hence, the correct answer is option C.

38. Questions

Answer: A

$$400 - 2^2 = 396$$

$$396 - 3^2 = 387 \text{ (Therefore, 389 is wrong and should be 387)}$$

$$387 - 5^2 = 362$$

$$362 - 7^2 = 313$$

$$313 - 11^2 = 192$$

Logic: The series follows the subtraction of consecutive primes squares.

Hence, the correct answer is option A.

39. Questions

Answer: A

$$55 + 15 = 70$$

$$70 + 25 = 95$$

$95 + 35 = 130$ (Therefore, 135 is wrong and should be 130)

$$130 + 45 = 175$$

$$175 + 55 = 230$$

Hence, the correct answer is option A.

40. Questions

Answer: E

$$8 * 2 = 16$$

$$16 * 1.5 = 24$$

$24 * 2 = 48$ (Therefore, 50 is wrong and should be 48)

$$48 * 1.5 = 72$$

$$72 * 2 = 144$$

Logic: The series alternates between multiplying by 2 and multiplying by 1.5.

Hence, the correct answer is option E.

41. Questions

Answer: A

I). $x^2 - 21x + 104 = 0$

$$x^2 - 13x - 8x + 104 = 0$$

$$x(x - 13) - 8(x - 13) = 0$$

$$(x - 8)(x - 13) = 0$$

$$x = 8, 13$$

II). $y^2 - 27y + 182 = 0$

$$y^2 - 14y - 13y + 182 = 0$$

$$y(y - 14) - 13(y - 14) = 0$$

$$(y - 13)(y - 14) = 0$$

$$y = 13, 14$$

Comparison:

$$8 < 13$$

$$8 < 14$$

$$13 = 13$$

$$13 < 14$$

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Therefore, $x \leq y$.

Hence, the correct answer is option A.

42. Questions

Answer: B

I). $2x^2 + 15x + 27 = 0$

$$2x^2 + 9x + 6x + 27 = 0$$

$$x(2x + 9) + 3(2x + 9) = 0$$

$$(x + 3)(2x + 9) = 0$$

$$x = -3, -4.5$$

II). $2y^2 + 19y + 45 = 0$

$$2y^2 + 10y + 9y + 45 = 0$$

$$2y(y + 5) + 9(y + 5) = 0$$

$$(2y + 9)(y + 5) = 0$$

$$y = -5, -4.5$$

Comparison:

$$-3 > -5$$

$$-3 > -4.5$$

$$-4.5 > -5$$

$$-4.5 = -4.5$$

Therefore, $x \geq y$.

Hence, the correct answer is option B.

43. Questions

Answer: E

I). $x^2 - 6x - 55 = 0$

$$x^2 - 11x + 5x - 55 = 0$$

$$x(x - 11) + 5(x - 11) = 0$$

$$(x + 5)(x - 11) = 0$$

$$x = -5, 11$$

II). $y^2 - 22y + 117 = 0$

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$$y^2 - 13y - 9y + 117 = 0$$

$$y(y - 13) - 9(y - 13) = 0$$

$$(y - 9)(y - 13) = 0$$

$$y = 9, 13$$

Comparison:

$$-5 < 9$$

$$-5 < 13$$

$$11 > 9$$

$$11 < 13$$

Therefore, the relationship cannot be established.

Hence, the correct answer is option E.

44. Questions

Answer: B

I). $x = \sqrt[3]{(784)}$

$$x = 28$$

II). $y^2 - 24y - 112 = 0$

$$y^2 - 28y + 4y - 112 = 0$$

$$y(y - 28) + 4(y - 28) = 0$$

$$(y + 4)(y - 28) = 0$$

$$y = -4, 28$$

Comparison:

$$28 > -4$$

$$28 = 28$$

Therefore, $x \geq y$.

Hence, the correct answer is option B.

45. Questions

Answer: C

I). $5x^2 - 18x + 16 = 0$

$$5x^2 - 10x - 8x + 16 = 0$$

$$5x(x - 2) - 8(x - 2) = 0$$

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$$(5x - 8)(x - 2) = 0$$

$$x = 1.6, 2$$

$$\text{II). } y^2 - 11y + 28 = 0$$

$$y^2 - 7y - 4y + 28 = 0$$

$$y(y - 7) - 4(y - 7) = 0$$

$$(y - 4)(y - 7) = 0$$

$$y = 4, 7$$

Comparison:

$$1.6 < 4$$

$$1.6 < 7$$

$$2 < 4$$

$$2 < 7$$

Therefore, $x < y$.

Hence, the correct answer is option C.

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